$$\begin{aligned}\Delta^4 y_3 &= (E-1)^4 y_3\\ &= (E^4-4E^3+6E^2-4E+1)y_3\\ &= E^4y_3-4E^3y_3+6E^2y_3-4Ey_3+y_3\\ &= y_7-4y_6+6y_5-4y_4+y_3\\ &= 17-4(9)+6(8)-4(-6)+2\\ &= 17-36+48+24+2=55\end{aligned}$$

5.1.2 Finding the missing terms

Using the difference operators and shifting operator we can able to find the missing terms.

Example 5.10

From the following table find the missing value

x	2	3	4	5	6
$f(x)$	45.0	49.2	54.1	-	67.4

Solution:

Since only four values of $f(x)$ are given, the polynomial which fits the data is of degree three. Hence fourth differences are zeros.

(ie) $\Delta^4 y_0 = 0,\ \therefore (E-1)^4 y_0 = 0$

$$(E^4-4E^3+6E^2-4E+1)y_0 = 0$$

$$E^4y_0-4E^3y_0+6E^2y_0-4Ey_0+y_0 = 0$$

$$y_4-4y_3+6y_2-4y_1+y_0 = 0$$

$$67.4-4y_3+6(54.1)-4(49.2)+45 = 0$$

$$240.2=4y_3 \quad \therefore y_3 = 60.05$$

Example 5.11

Estimate the production for 1964 and 1966 from the following data

Year	1961	1962	1963	1964	1965	1966	1967
Production	200	220	260	-	350	-	430

Solution:

Since five values are given, the polynomial which fits the data is of degree four.

Hence $\Delta^5 y_k = 0$ (ie) $(E-1)^5 y_k = 0$

i.e., $(E^5-5E^4+10E^3-10E^2+5E-1)y_k = 0$

$$E^5y_k-5E^4y_k+10E^3y_k-10E^2y_k+5Ey_k-y_k = 0 \quad (1)$$

Put $k=0$ in (1)

$$E^5y_0-5E^4y_0+10E^3y_0-10E^2y_0+5Ey_0-y_0 = 0$$

$$y_5-5y_4+10y_3-10y_2+5y_1-y_0 = 0$$

$$y_5-5(350)+10y_3-10(260)+5(220)-200 = 0$$

$$y_5+10y_3 = 3450 \quad (2)$$

Put $k=1$ in (1)

$$E^5y_1-5E^4y_1+10E^3y_1-10E^3y_1-y_1 = 0$$

$$y_6-5y_5+10y_4-10y_3-y_1 = 0$$

$$430-5y_5+10(350)-10y_3+5(260)-220 = 0$$

$$5y_5+10y_3 = 5010 \quad (3)$$

(3) – (2) $\Rightarrow$ $4y_5 = 1560$

$$y_5 = 390$$

From (1) $390+10y_3 = 3450$

$$10y_3 = 3450-390$$

$$y_3 \cong 306$$

			1			
		1	2	1		
	1	3	3	1		
1	4	6	4	1		
1	5	10	10	5	1	

Exercise 5.1

1. Evaluate $\Delta(\log ax)$.
2. If $y = x^3-x^2+x-1$ calculate the values of y for $x = 0,1,2,3,4,5$ and form the forward differences table.
3. If $h = 1$ then prove that $(E^{-1}\Delta)x^3 = 3x^2-3x+1$.

XII Std - Business Maths & Stat EM Chapter 5.indd 111 2/28/2020 6:09:11 PM